Performance Improvements for OpenFOAM
We make your cases run faster

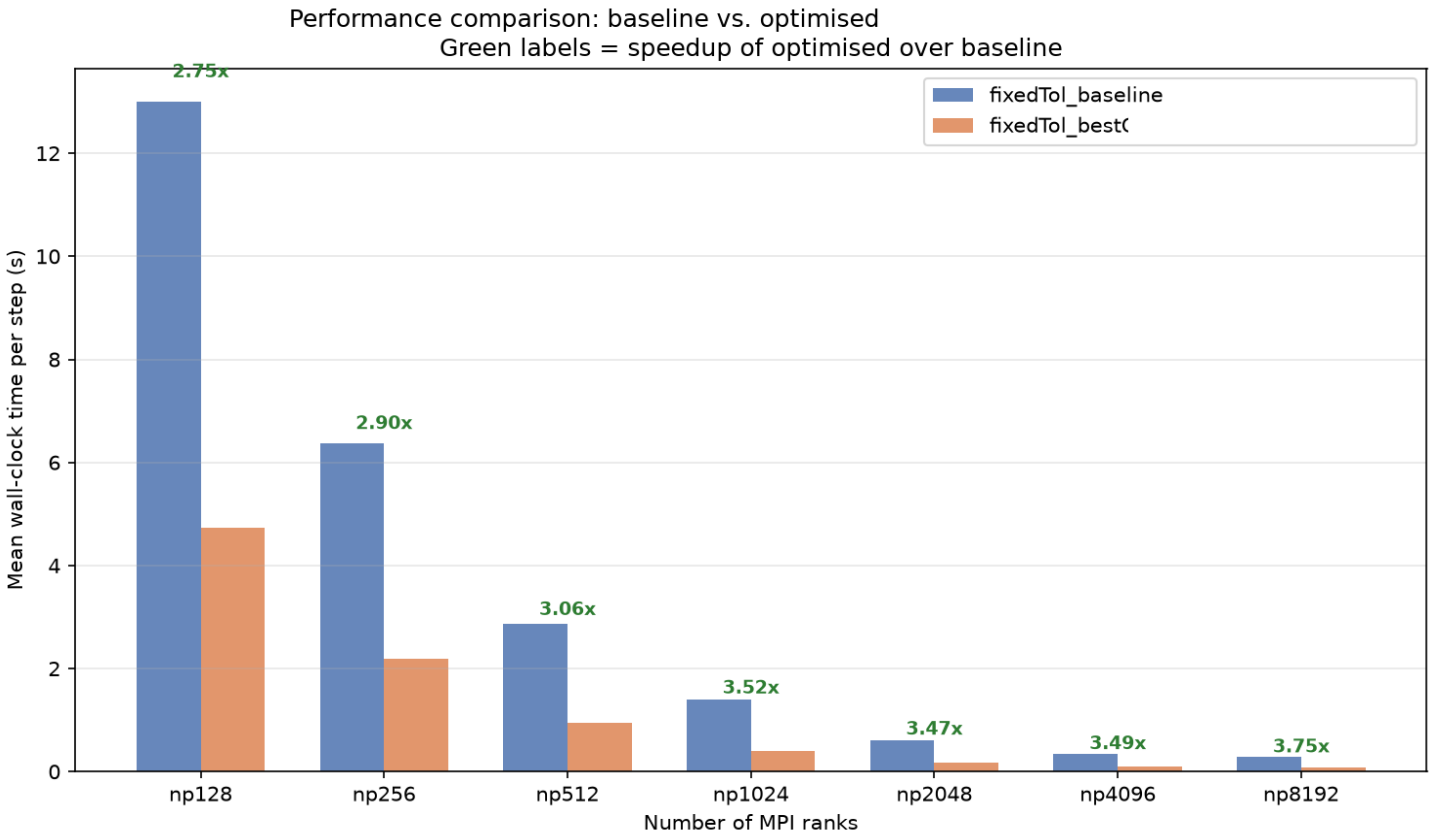
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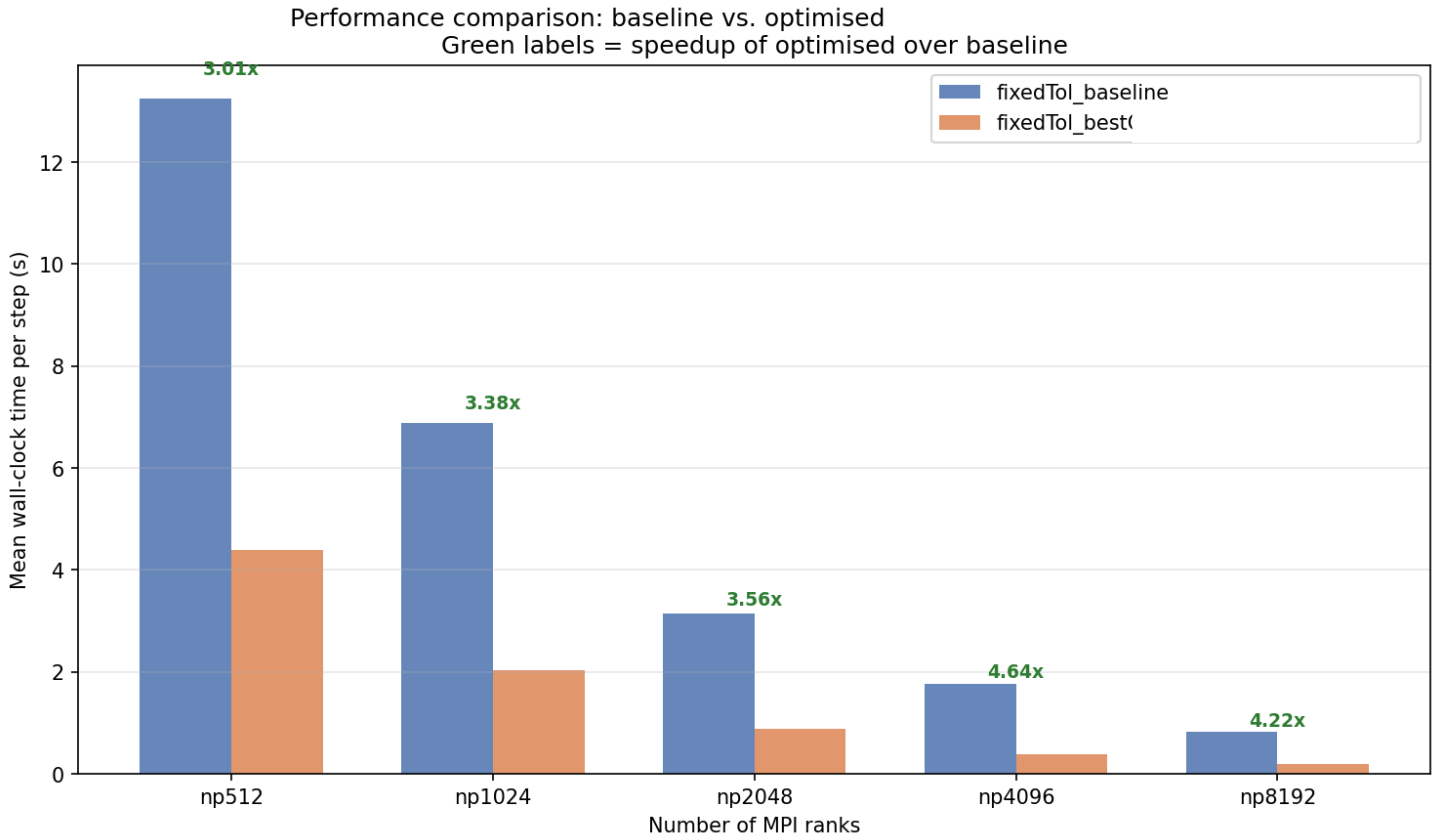
HPC activities at Wikki

- Wikki has extensive expertise in tuning OpenFOAM in HPC environments
  - Partner in Inno4Scale innovation studies (asynchronous linear solvers and asynchronous particle tracking)
  - Partner and work-package leader in exaFOAM project
    - Collection and publication of HPC benchmark cases
    - Award-winning new parallel I/O (coherent mesh format)
    - Parallel load balancing
  - Organisation of and participation in the OpenFOAM HPC Challenge demonstrating up to x2.0 overall speed-up
- Extensive commercial expertise through support contracts
- Independent of specific hardware or software (OF flavour)

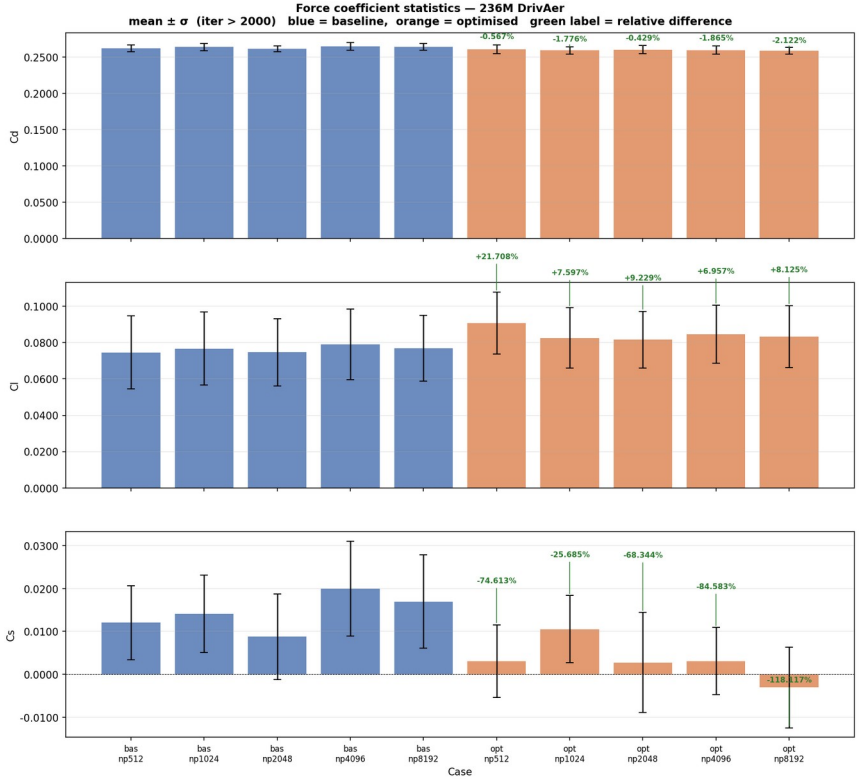
Performance comparison for 65M mesh



Performance comparison for 236M mesh



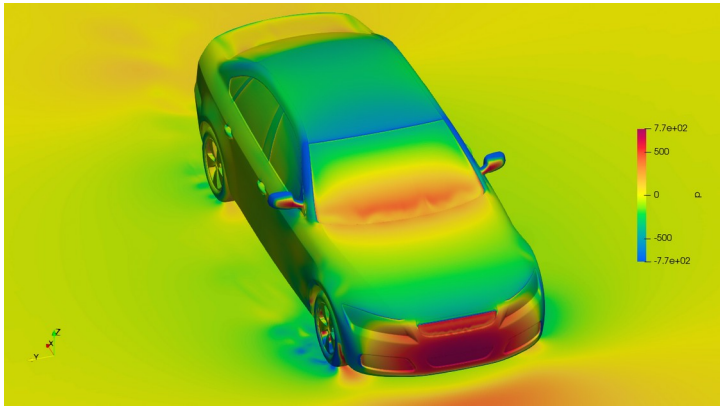
Performance comparison for 236M mesh



Our results presented for OpenFOAM HPC Challenge 2

Speedup w.r.t. to baseline

N partitions	Mesh size	
	65M	236M
128	x2.75	
256	x2.90	
512	x3.06	x3.01
1024	X3.52	x3.38
2048	X3.47	x3.56
4096	X3.49	x4.64
8192	X3.75	x4.22



- DrivAer: external car aerodynamics, simpleFoam
- https://develop.openfoam.com/committees/hpc/-/tree/develop/incompressible/simpleFoam/occDrivAerStaticMesh

Our methodology for better performance

- Review the case setup and apply our best-practices
- Detailed analysis using improved built-in profiling, Score-P, Extrae, likwid, perf, … Analysis of assembly code
- Implement/integrate/deploy performance-enhancing software components
  - New algorithmns
  - New datastructures
  - Use of modern MPI features
- Quantify performance (improvements) using statistical methods
- Validate improved methodology against acceptance criteria provided by the the Client

Our ingredients for better performance

- Optimised settings (system/fvSchemes, system/fvSolution, ...)
- Better utilisation of memory bandwidth
  - Improved computational kernels
  - Cache-aware matrix renumberings
- Improved parallel efficiency
  - Topology-aware mesh partitioners
  - Non-blocking asynchronous linear solvers
- Hardware-specific optimisations
- Parallel and/or asynchronous I/O
- Parallel load balancing on various levels
- Asynchronous GPU offloading of selected functionality
- Use of optimisation loops and AI

Business Proposition

Wikki analyses the Client’s case and demonstrates speed-ups for free

- Client provides:
  - baseline case and execution workflow
  - HPC specification (target platform, number of cores, …)
  - Acceptance criteria and associated automatic evaluation
  - Optionally: Access to hardware for running the benchmarks
  - Optionally: Non-binding indication of potential budget (X1 EUR for S1 speed-up; X2 EUR for S2 speed-up; …)
- Wikki provides:
  - Overall speed-up and information about the ingredients used to achieve it